UserService

```
public Res<UserSL> signIn(un, pass) {
  try {
     UserBL ubl = uf.signin(un, pass);
     UserSL usl = new UserSL(ubl)
      return new Res(usl, null)
  } catch (e) {
     return new Res(null, "Login failed bad us/pass")
  }
}


///<summary>
///this method entring the user to the system
///<param "email"> the email address of the user that log in
///<returns> String JSON, unless an error occurs when one of the inputs is
wrong according to the data in the program and return an appropriate error
massege
public String SignIn(String email, String password)
{
    return null;
}

///<summary>
///this method create a new user
///<param email> the email of the new user
///<param "username"> the username of the new user
///<param password> the password that a new user choose
///<returns> String JSON, unless an error occurs when one of the inputs is
wrong and return an appropriate error massege
public String SignUp(String email, String username, String password)
{
    return null;
}

///<summay>
///this method present if the user is in the system
/// <returns> true if the user is connected and false if he is not
public bool IsLoggedIn(string Username)
{
    return null;
}
```

BoardService

```
/// this method creates a new board.
/// param actor ="user" is the unique name of the user
/// param String="title" of the board
/// returns String JSON
/// returns an error if the wanted name is taken.
public String CreateBoard (String title)
{
   return null;
}


/// this method gets as input an existing board and it's creator
/// deletes this board and all its tasks
/// changes the pointers users have to the board to null.

public  void DeleteBoard( long boardID)
{
   return null;
}


/// param "newlimit" = the restriction of the wanted status column.
/// param " column" = the status between the three: TO DO, IN PROGGRESS,
DONE.
/// this method changes the restriction on the amount of tasks in each status.
/// returns an error if the newlimit is smaller then the amount of tasks in the specific
column.
public int Change_Limit(int newlimit, int column, long boardID){
   return null;
}


///<returns> string JSON
/// this method goes in loop through all boards and enters a them to a list.
public String presentBoards( String email) {
   return null;
}


///<returns> String JSON
///this method creates a pointers to the users that have been shared with the board.
public String Share(String email ,long boardID) {
   return null;
}
```

TaskService

```
///<summary>
///this method create a new task
///<param title> the tite of the task
///<param description> the description of the task
///<param deadline> the last date to do the task
///<returns> string JSON. appears in the appropriate board.
return an error massege if the date is invalid or the string is
empty
public String CreateTask(String title, String description,
Datetime deadline)
{
   return null;
}

///<summary>
///this method moving the task to the next column
///<param TID> the unique ID of each task
//<returns>string JSON
public String  MoveForward (long taskID)
{
      return null;
}


///<summary>
///this method editing the task
///<param title> the new title of the task
///<param TID> the unique ID of the task
///<param deadline> the new deadline to the task
///<param description> new description of the task
///<returns> string JSON the update information about the task,
by default the parameters stay the same as they were unless
the user is change them. error messege will send according to
the same conditions in thr CreateTask method

public String EditTask ( String title, long BID, Datetine
deadline, String description)
{
   return null;
}
```

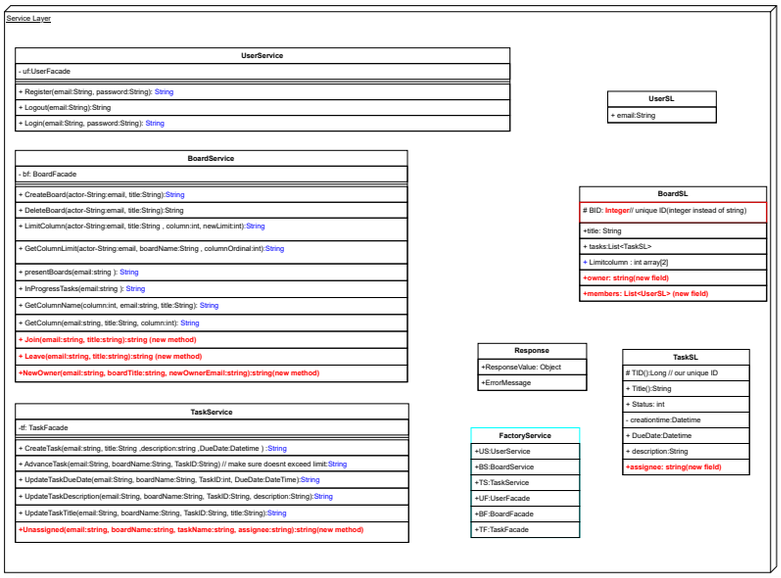
BoardService

```
Response<BoardSL> CreateBoard (username,title)
{
try{
BoardBL bbl = bf.createBoard(username,title)
BoardSL bsl = new BoardSL(bbl)
return new response(bsl,null)

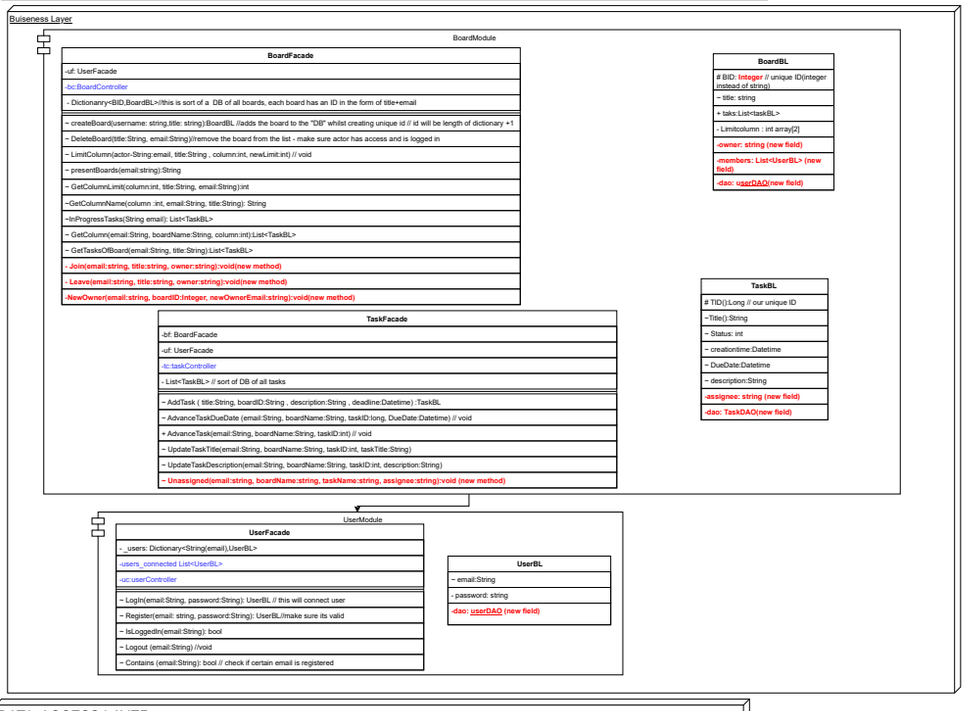
}
(Exception e) {
    return new Res(null, e.Message)
  }
}


public Response<BoardSL>
```

Service Layer



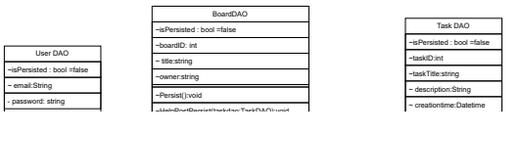
Buiseness Layer

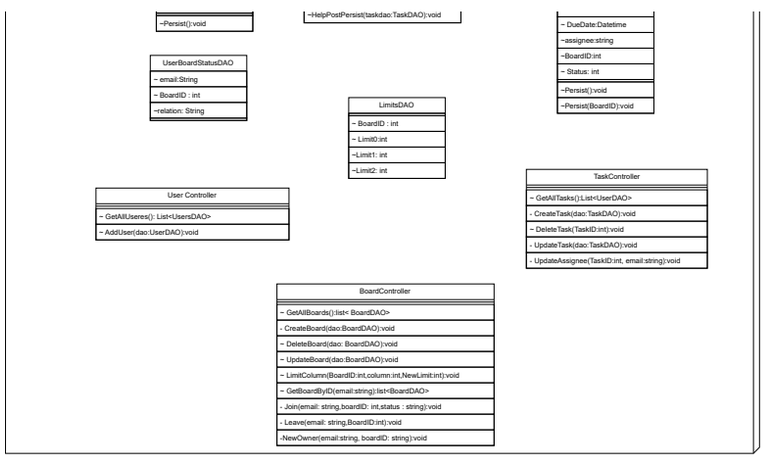



```
DeleteBoard(username: string, title:string)
{
 List<BoardBL>.remove(title+username)
}
```

```
PresentBoard(email:string)
{
 UserBL user=Member.get(email)
  if(UserBL!=isEmpty){
      Iterator iter=user.getBoardList.iterator()
      while(iter.hasNext()){
          if(iter.next==null){List<BoardBL>.remove(iter)}
          else{Print(List<BoardBL>.get(iter)}
      }
}
```

DATA ACCESS LAYER





template	
1	cdvdascsd
2	sdcdcsdc
3	sdcsdcds
1	Value 1
1	Value 1
2	Value 2
3	Value 3
2	Value 2
3	Value 3